

Comparing Novel Thermodynamic Computing with Traditional Graphics Processing Unit Implementations for Ground State Energy Estimation in Classical Optimization Problems

Noah Kestenholz

Department of Advanced Computing Sciences

Faculty of Science and Engineering

Maastricht University

Maastricht, The Netherlands

Abstract—Thermodynamic computing harnesses thermal noise as a computational resource, with block Gibbs sampling as its algorithmic core. The THRML library, developed by Extropic for AI applications such as diffusion-like models and energy-based models, simulates this approach on conventional GPU hardware. This thesis evaluates whether THRML’s rejection-free block Gibbs transfers effectively to classical combinatorial optimisation, benchmarking it against Metropolis-Hastings (MH) and standard single-site Gibbs on GPU. We benchmark all three samplers across ferromagnetic Ising, anti-ferromagnetic Ising, and MaxCut instances on 2D square lattices (up to 100×100 nodes) and random regular graphs (up to 2,500 nodes), with a sweep over inverse temperature β . Statistical comparisons use the Mann-Whitney U test on 150 pooled per-chain best values across three independent runs. THRML achieves two to three orders of magnitude higher effective samples per second (ESS/s) than MH and standard Gibbs across all configurations, while matching or exceeding MH in solution quality on the majority of problem instances. The quality advantage is most pronounced on MaxCut degree-5, where effect sizes grow monotonically with problem size (rank-biserial r up to 0.34 vs. standard Gibbs at 2,500 nodes). The sole exception is the very low temperature regime ($\beta = 2.0$), where MH’s rejection step yields a small but significant quality advantage. These results support the algorithmic premise of thermodynamic sampling hardware and identify MaxCut on dense graphs as the problem class where block Gibbs provides the greatest benefit over single-site methods.

Index Terms—thermodynamic computing, THRML, Ising model, MaxCut, Gibbs sampling, Block Gibbs, Metropolis-Hastings, probabilistic computing, energy-based models, EBMs

I. INTRODUCTION

Solving hard combinatorial optimisation problems is central to applications spanning logistics, drug discovery, and machine learning, yet the energy cost of running these workloads on conventional hardware continues to grow. Thermodynamic computing has emerged as a promising alternative paradigm: rather than suppressing thermal noise, it harnesses it as a computational resource [7], [8]. Here we study this paradigm

at the software level, benchmarking GPU implementations of block and single-site MCMC samplers on canonical discrete optimisation problems.

The Ising model is the canonical test bed for combinatorial solvers. It assigns a binary spin $s_i \in \{-1, +1\}$ to each node of a graph, with an energy function whose minimisation is NP-hard in general. Any NP problem can be mapped to an Ising instance with only polynomial overhead [1], making progress on Ising solvers broadly significant. This thesis studies two regimes: the *ferromagnetic* case ($J_{ij} > 0$), where aligned spins are favoured and the energy landscape is relatively smooth, and the *anti-ferromagnetic* case ($J_{ij} < 0$), which produces frustrated, rugged landscapes. The MaxCut problem—partition a graph to maximise the number of crossing edges—maps exactly to anti-ferromagnetic Ising and serves as the natural stress test for any sampler [11].

The standard baseline for ground state estimation is Metropolis-Hastings (MH), whose rejection step allows the Markov chain to escape local minima and guarantees convergence to the target distribution. GPU implementations exploit the local neighbourhood structure of spin models to achieve speed-ups of up to three orders of magnitude over serial CPU code [2], [3]. Gibbs sampling eliminates rejection by sampling each spin directly from its conditional distribution, removing the accept/reject overhead but preventing the chain from crossing low-probability barriers—a failure mode that is particularly severe in frustrated anti-ferromagnetic systems [4]. Standard single-site Gibbs serves as a natural intermediate baseline: it shares the rejection-free property of block Gibbs but without parallelism, isolating the contribution of block updates. Block Gibbs (THRML) extends this by using graph colouring to identify conditionally independent spins that can be updated simultaneously as a block, gaining throughput over single-site Gibbs while retaining this fundamental limitation.

Extropic’s Thermodynamic Sampling Unit (TSU) is built around block Gibbs, deliberately forgoing rejection in exchange for the energy efficiency that probabilistic hardware enables. Jelinčič et al. project performance parity with GPUs

This thesis was prepared in partial fulfilment of the requirements for the Degree of Bachelor of Computer Science, Maastricht University. Supervisors: Georgios Stamoulis and Charis Kouzinopoulos.

at roughly 10,000 times lower energy [7], building on earlier work on p-bit circuits [5] and sparse Ising Machines [6]. The THRML library [9] simulates TSU-style block Gibbs on a conventional GPU, primarily targeting AI workloads such as energy-based models and diffusion-like generative models. This thesis applies THRML to classical combinatorial optimisation—Ising ground-state estimation and MaxCut—to evaluate how well its block Gibbs approach transfers beyond its intended AI domain.

The broader landscape of Ising solvers spans a wide range of hardware paradigms [1]. Quantum annealers—most notably D-Wave systems—exploit quantum tunnelling to escape local minima, but are currently limited to sparse, hardware-native graph topologies and relatively small qubit counts. Coherent Ising Machines based on optical parametric oscillators can tackle dense instances at optical speeds, but require specialised photonic hardware. On the probabilistic hardware side, p-bit circuits [5] and sparse Ising Machines [6] implement rejection-free Boltzmann sampling with stochastic binary units, sharing the core philosophy of block Gibbs. Among purely software approaches, GPU-accelerated MCMC [2], [3] represents the accessible state-of-the-art: it exploits the local neighbourhood structure of spin models to parallelise proposals across thousands of CUDA cores. THRML bridges these paradigms by simulating probabilistic hardware-style block Gibbs on a conventional GPU, making it directly comparable to GPU MCMC baselines without requiring specialised hardware.

Despite these advances, the empirical performance trade-offs of block Gibbs across different problem classes remain incompletely characterised across practical problem scales. It is unclear whether THRML block Gibbs achieves solution quality comparable to MH, and under which conditions its computational efficiency advantage holds. This thesis addresses that gap through a systematic three-way comparison of THRML block Gibbs, standard single-site Gibbs, and Metropolis-Hastings, all running on identical GPU hardware. Experiments cover two graph topologies—2D square lattice and random regular graph (RRG)—across ferromagnetic Ising, anti-ferromagnetic Ising, and MaxCut instances. Grid sizes range from 50×50 to 100×100 (up to 10,000 nodes); RRG sizes range from 900 to 2,500 nodes. MaxCut is evaluated at graph degrees 3 and 5 to test the effect of connectivity. An additional sweep over inverse temperature $\beta \in \{0.1, 0.5, 1.0, 2.0\}$ on the 70×70 grid probes sampler behaviour at low and high temperatures.

Three research questions guide the work:

- 1) How do the three samplers compare in solution quality across problem types and system sizes?
- 2) How does their relative performance scale with graph topology and temperature?
- 3) What are the computational efficiency trade-offs, as measured by effective samples per second and Gelman–Rubin convergence diagnostics?

The remainder of this thesis is structured as follows: Section II describes the three sampler implementations; Section III defines the experimental methodology; Section IV presents

results; Section V interprets findings in the context of existing literature; and Section VI draws conclusions.

II. METHODS

A. Problem Formulation

All three samplers operate on the same shared graph representation. An Ising model instance is defined by a graph $G = (V, E)$ with $n = |V|$ nodes, edge-wise coupling strengths $J_{ij} \in \mathbb{R}$, and inverse temperature $\beta > 0$. Each node i carries a spin $s_i \in \{-1, +1\}$, and the energy of a configuration $\mathbf{s}$ is:

$$\mathcal{E}(\mathbf{s}) = -\beta \sum_{(i,j) \in E} J_{ij} s_i s_j \quad (1)$$

The samplers target the Boltzmann distribution $P(\mathbf{s}) \propto \exp(-\mathcal{E}(\mathbf{s}))$. For ferromagnetic instances $J_{ij} = +1$, aligned spins are energetically favoured; for anti-ferromagnetic instances $J_{ij} = -1$, opposing spins are favoured. MaxCut is handled through the same formulation by setting $J_{ij} = -1$, so that minimising (1) is equivalent to maximising the number of cut edges. The cut value is computed separately as $\text{Cut}(\mathbf{s}) = \sum_{(i,j) \in E} |J_{ij}| (1 - s_i s_j)/2$, which for unit weights ($|J_{ij}| = 1$) counts the number of cut edges.

B. Metropolis-Hastings

At each step, a node i is chosen uniformly at random and a spin flip is proposed. The energy change from flipping s_i depends only on the local neighbourhood $\mathcal{N}(i)$:

$$\Delta\mathcal{E} = 2\beta s_i \sum_{j \in \mathcal{N}(i)} J_{ij} s_j \quad (2)$$

The flip is accepted with probability $\min(1, \exp(-\Delta\mathcal{E}))$. One sweep consists of n such steps. The acceptance step guarantees detailed balance and allows the chain to cross low-probability barriers, at the cost of rejected proposals that leave the state unchanged.

The implementation uses a padded neighbour list of shape $[n, d_{\max}]$, where $d_{\max}$ is the maximum node degree, with a boolean mask to handle varying degree. The inner loop over n steps is compiled with `jax.lax.scan`; PRNG keys are derived on-the-fly via `jax.random.fold_in` to avoid materialising large key arrays in GPU memory. The sampler is JIT-compiled with `jax.jit` and parallelised over independent chains with `jax.vmap` [10].

C. Standard Gibbs Sampling

Gibbs sampling draws each spin exactly from its conditional distribution given all neighbours, eliminating the accept/reject step entirely. For node i :

$$P(s_i = +1 \mid \mathbf{s}_{\setminus i}) = \sigma \left(2\beta \sum_{j \in \mathcal{N}(i)} J_{ij} s_j \right) \quad (3)$$

where σ is the sigmoid function. One sweep consists of n single-node conditional updates. The implementation shares

the same neighbour list structure and `lax.scan/vmap` pattern as MH; the only difference is replacing the Metropolis accept/reject with direct conditional sampling from (3).

D. Block Gibbs via THRML

Block Gibbs extends single-site Gibbs by updating entire groups of conditionally independent nodes in parallel. Two nodes are conditionally independent given all others if and only if they share no edge, so the natural grouping is a proper graph colouring: nodes of the same colour can be sampled simultaneously without violating the Markov property.

The graph is coloured using the DSATUR greedy algorithm [13], producing colour classes $C_1, \dots, C_k$. One block Gibbs step cycles through all classes in order; within each class, all spins are sampled in parallel from their conditionals (3). A 2D square grid is bipartite and requires two colours; a degree-3 random regular graph typically requires three or four. More colours mean smaller parallel blocks but the same total work per step.

This sampler is implemented using the THRML library [9], which provides an `IsingEBM` energy model, an `IsingSamplingProgram` encoding the colour-block structure, and a `sample_states` function that executes the schedule. The same `jax.vmap` parallelism over independent chains is applied externally.

E. Shared Infrastructure and Timing

All three samplers share a common warm-up phase of n_{warmup} sweeps that are discarded before sample collection. Samples are then collected every n_{thinning} sweeps to reduce autocorrelation. Wall-clock time is measured using `jax.block_until_ready` to ensure all GPU computation is complete before the timer stops, and a separate untimed compilation run precedes each timed run to exclude JIT overhead. This ensures wall-clock times reflect steady-state throughput and are directly comparable across samplers.

III. EXPERIMENTS

The experiments are designed to answer the three research questions by varying problem type, graph topology, system size, and temperature independently, while holding all sampler hyperparameters fixed across conditions. This section describes the experimental design, the configurations tested, and the metrics collected.

A. Experimental Design

Each experiment runs all three samplers—MH, standard Gibbs, and THRML block Gibbs—on the same problem instance with independently sampled random initialisations, chain count, and sample budget. This ensures differences in outcome reflect algorithmic behaviour rather than initialisation or resource differences.

All experiments were run on a single laptop NVIDIA RTX 3060 (6 GB VRAM). Each sampler was given $n_{\text{chains}} = 50$ independent parallel chains, initialised from random spin configurations. Each chain ran $n_{\text{warmup}} = 200$ warm-up sweeps

(discarded), then collected $n_{\text{samples}} = 500$ samples with $n_{\text{thinning}} = 5$ sweeps between each sample. Each configuration was repeated for $N_{\text{runs}} = 3$ independent runs; chains are pooled across runs (150 total per sampler) for statistical analysis. JIT compilation time was excluded from all timing measurements.

B. Problem Instances and Configurations

Four groups of configurations were benchmarked, covering the three problem types and three graph topologies.

1) *Ferromagnetic and Anti-ferromagnetic Grid*: A 2D square lattice with $J = +1$ (ferromagnetic) and $J = -1$ (anti-ferromagnetic) at $\beta = 1.0$. System sizes of 50×50 , 70×70 , and 100×100 (2,500 to 10,000 nodes) are tested to measure scaling behaviour; these sizes were chosen to span a range where critical slowing down becomes progressively more severe, while remaining within the 6 GB VRAM budget when running 50 chains in parallel. The 2D grid is bipartite, so THRML requires only two colour classes regardless of size.

2) *Temperature Sweep*: A ferromagnetic grid ($J = +1$) at inverse temperatures $\beta \in \{0.1, 0.5, 1.0, 2.0\}$ on a fixed 70×70 lattice, sweeping from the high-temperature disordered phase ($\beta = 0.1$) to the strongly ordered phase ($\beta = 2.0$). The 70×70 size was chosen as a middle ground large enough to make the ordered phase hard to mix on, revealing whether THRML's throughput advantage holds across the full temperature range rather than at a single operating point.

3) *Random Regular Graph*: A random regular graph of degree $d = 3$ with ferromagnetic coupling $J = +1$ at $\beta = 1.0$. Node counts of 900, 1,600, and 2,500 are tested to measure scaling behaviour; these sizes were chosen to span a comparable node-count range to the grid experiments under the same VRAM budget. Unlike the grid, a degree-3 RRG is not bipartite and typically requires three or four colour classes under DSATUR.

4) *MaxCut*: MaxCut on a random regular graph ($J = -1$) at $\beta = 1.0$, evaluated at degrees $d \in \{3, 5\}$ and node counts of 900, 1,600, and 2,500. The degree sweep isolates the effect of graph connectivity on block Gibbs parallelism: higher degree increases the chromatic number, reducing colour-class size and therefore the parallelism available to THRML.

C. Metrics

Four metrics are recorded for each configuration and sampler:

- **Solution quality**: the mean best energy (or cut value for MaxCut) found across chains, and the distribution of per-chain best values. Statistical significance of differences between samplers is assessed using the Mann-Whitney U test on the 150 pooled per-chain best values ($3 \text{ runs} \times 50 \text{ chains}$), with effect size reported as the rank-biserial correlation r .
- **Convergence**: the Gelman-Rubin $\hat{R}$ statistic [12] computed on the energy trajectory across all 50 chains. Values $\hat{R} < 1.1$ indicate convergence; values substantially above 1.1 indicate chains have not mixed.

- **Sampling efficiency:** the effective sample size (ESS) per chain, estimated via integrated autocorrelation time using the FFT [14], and ESS per second as the primary throughput metric.
- **Wall-clock time:** total wall time for the timed run, excluding JIT compilation.

IV. RESULTS

Results are reported across four experiment groups: grid ferromagnetic, grid anti-ferromagnetic, ferromagnetic random regular graph (RRG), and MaxCut on RRG at degrees 3 and 5. For each group, solution quality (mean best energy or cut value), convergence diagnostics ($\hat{R}$), and sampling efficiency (ESS/s) are reported. Statistical comparisons use the Mann-Whitney U test on 150 pooled per-chain best values (3 runs $\times$ 50 chains); effect sizes are reported as rank-biserial correlation r .

A. Grid: Ferromagnetic and Anti-ferromagnetic

Table I summarises the statistical comparisons. On ferromagnetic grids, solution quality differences between THRML and MH are not significant at any size ($p > 0.25$). Both THRML and MH significantly outperform standard Gibbs at 70×70 and 100×100 ($p < 0.0001$, $|r| = 0.30\text{--}0.48$), with the gap widening with system size. On anti-ferromagnetic grids the same pattern holds, with the additional result that at 100×100 THRML significantly outperforms MH ($p = 0.0012$, $r = -0.214$, **).

Gelman-Rubin diagnostics reveal that all three samplers fail to converge within the sample budget on grid instances: $\hat{R}$ ranges from 1.18 to 1.82 across all grid configurations (Figure 1). Figure 2 shows the energy trace for the 100×100 ferromagnetic grid: all three samplers are still descending at the end of the sample window, confirming incomplete mixing. Despite this shared failure to converge, sampling efficiency differs by orders of magnitude: at 50×50 , THRML achieves 604.7 ESS/s compared to 0.8 for MH and 0.1 for standard Gibbs—a $756\times$ gap over MH. At 100×100 , THRML achieves 13.8 ESS/s while both MH and Gibbs fall below 0.05 ESS/s. Wall-clock time scales steeply for MH and Gibbs as system size grows (Figure 3), while THRML’s block parallelism partially offsets this cost.

B. Temperature Sweep

At high temperature ($\beta = 0.1$), all three samplers converge fully ($\hat{R} = 1.000$) and produce statistically equivalent solution quality ($p > 0.06$). THRML achieves 3,462 ESS/s at $\beta = 0.1$ compared to 2.9 for MH and 2.8 for standard Gibbs—a $1,194\times$ gap. As temperature decreases, $\hat{R}$ rises for all samplers and ESS/s falls. At $\beta = 2.0$, MH significantly outperforms THRML in solution quality ($p = 0.028$, $r = 0.125$, *), while both outperform standard Gibbs ($p < 0.003$). Figure 4 shows solution quality and wall-clock time as functions of β .

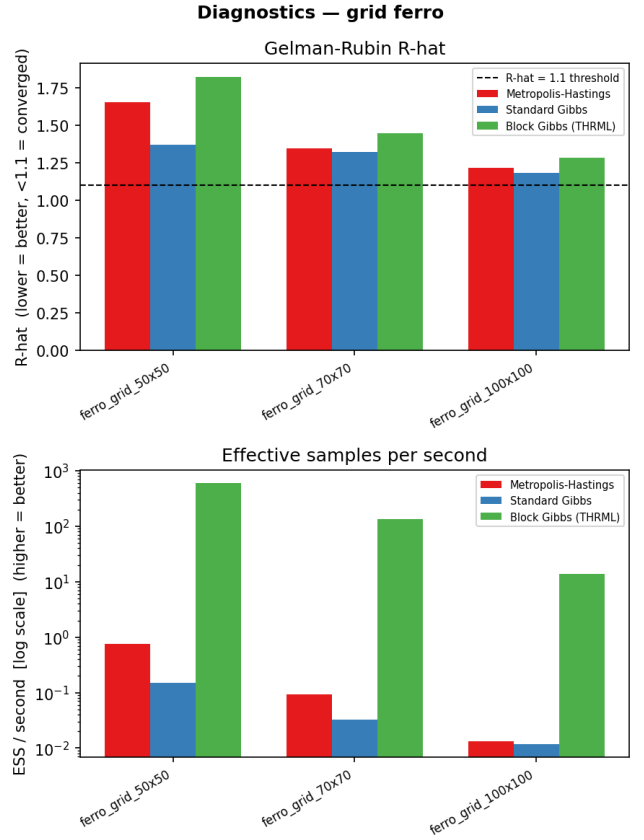


Fig. 1. Gelman-Rubin $\hat{R}$ (top) and ESS/s on log scale (bottom) for the ferromagnetic grid across system sizes. Dashed line indicates the $\hat{R} = 1.1$ convergence threshold.

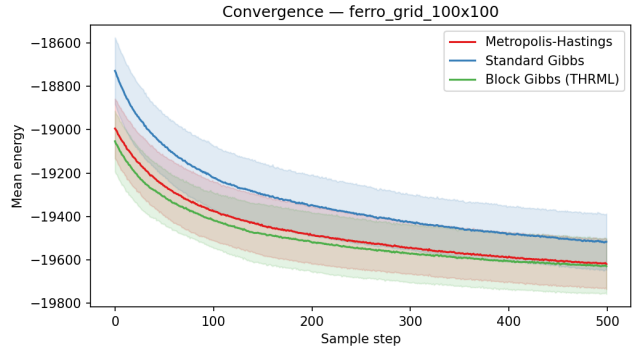


Fig. 2. Mean energy trace (± 1 std across chains) for the 100×100 ferromagnetic grid. All three samplers continue to descend throughout the sample window, indicating incomplete mixing within the sample budget.

C. Random Regular Graph

All three samplers converge fully on ferromagnetic RRG instances: $\hat{R} \leq 1.001$ across all sizes, providing a clean baseline for comparing throughput. At 900 nodes all samplers find identical mean best energy (-1350.00 , $p = 1.0$). At 1,600 nodes THRML marginally but significantly outperforms both MH and Gibbs ($p = 0.032$, $r = -0.040$, *; $p = 0.003$, $r = -0.067$, **); at 2,500 nodes differences are not signifi-

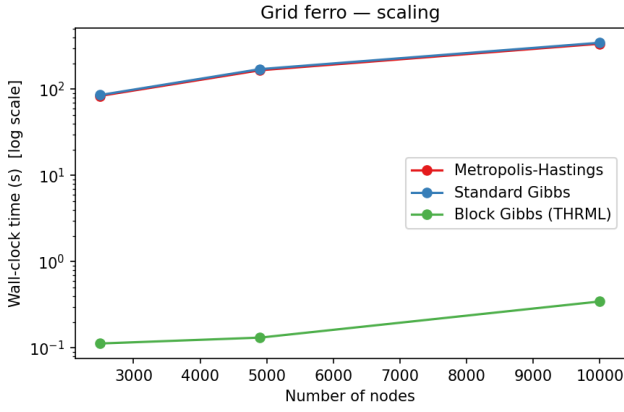


Fig. 3. Wall-clock time vs. number of nodes on log scale for the ferromagnetic grid. Each point is the mean across 3 runs.

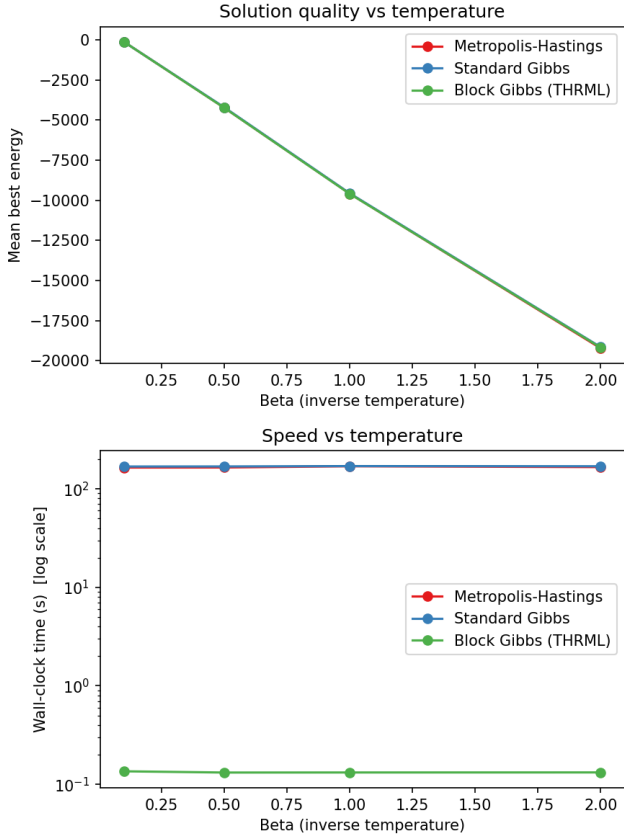


Fig. 4. Mean best energy (top) and wall-clock time on log scale (bottom) vs. inverse temperature β on a 70×70 ferromagnetic grid.

cant. THRML ESS/s is $325\times$ higher than MH at 900 nodes (4,527 vs. 13.9) and $781\times$ higher at 2,500 nodes (3,904 vs. 5.0), with all three samplers producing statistically equivalent solution quality at all sizes. Figure 5 shows how ESS/s scales with system size for each sampler.

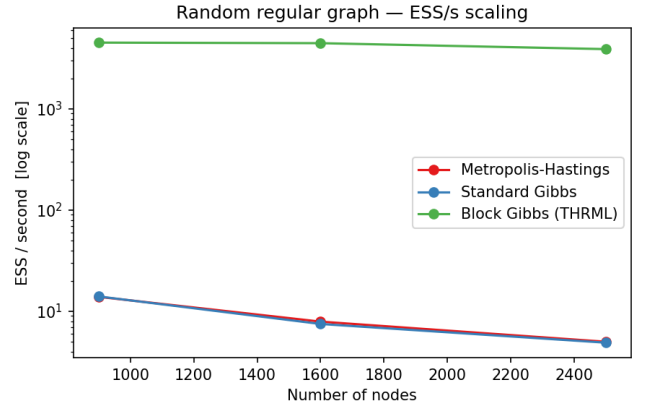


Fig. 5. ESS/s vs. number of nodes on log scale for the ferromagnetic RRG. THRML maintains two to three orders of magnitude higher throughput than MH and standard Gibbs at all sizes, while solution quality is statistically equivalent.

D. MaxCut

1) *Degree 3*: $\hat{R}$ values remain below 1.04 for all samplers at all sizes, indicating near-convergence. THRML achieves higher mean best cut value at 900 nodes ($p = 0.027$, $r = 0.147$, *) and 2,500 nodes ($p = 0.003$, $r = 0.199$, **; vs. Gibbs $p = 0.0009$, $r = 0.222$, * * *); differences are not significant at 1,600 nodes. MH and standard Gibbs are statistically equivalent at all sizes. THRML ESS/s exceeds that of MH by factors of 470 to 1,060 across the three sizes.

2) *Degree 5*: MaxCut degree-5 produces the strongest and most consistent quality advantage for THRML (Figure 7). THRML significantly outperforms MH at all three sizes ($p = 0.013$, $r = 0.165$ at 900 nodes; $p = 0.012$, $r = 0.168$ at 1,600 nodes; $p = 0.0001$, $r = 0.265$ at 2,500 nodes). THRML also outperforms standard Gibbs at 1,600 and 2,500 nodes ($p = 0.038$, $r = 0.138$; $p < 0.0001$, $r = 0.336$). Effect sizes grow monotonically with problem size for both comparisons: vs. MH, r increases from 0.165 to 0.168 to 0.265; vs. Gibbs, from 0.094 to 0.138 to 0.336. $\hat{R}$ remains below 1.09 for all samplers. Figure 6 shows ESS/s: THRML achieves 671, 386, and 229 ESS/s at sizes 900, 1,600, and 2,500 respectively, compared to below 2.0 for MH and Gibbs at all sizes—a gap of $115\times$ to $350\times$.

V. DISCUSSION

A. Sampling Efficiency

The most consistent finding across all experiment groups is that THRML achieves dramatically higher ESS/s than both MH and standard Gibbs—typically two to three orders of magnitude. This is not a consequence of faster wall-clock time per step; rather, THRML’s block updates decorrelate the chain more rapidly, producing more independent samples per unit time. This result is consistent with the theoretical motivation for block Gibbs: updating entire colour classes simultaneously breaks the local correlations that cause single-site samplers to mix slowly, particularly on structured graphs [4].

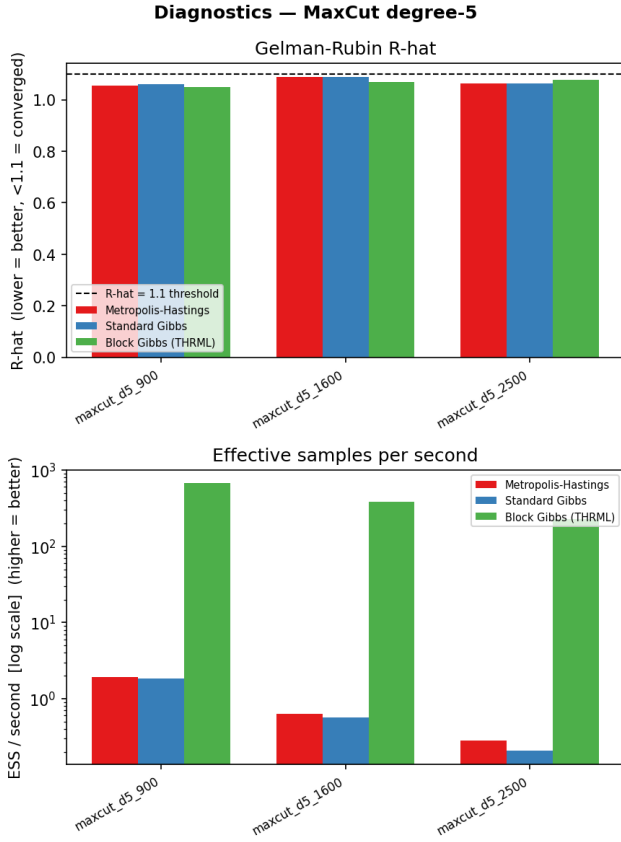


Fig. 6. Gelman-Rubin $\hat{R}$ (top) and ESS/s on log scale (bottom) for MaxCut degree-5 across system sizes.

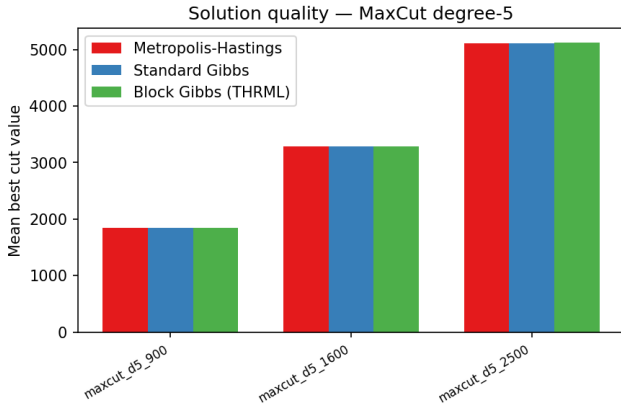


Fig. 7. Mean best cut value across samplers for MaxCut degree-5. Higher is better. Error bars show standard deviation across 150 pooled chains.

Weigel et al. demonstrated GPU speed-ups of up to three orders of magnitude for MH over serial CPU implementations by exploiting the checkerboard decomposition of the 2D lattice [2], [3]. Our results show that THRML’s generalised graph colouring achieves a further two to three orders of magnitude improvement in ESS/s over GPU-accelerated MH, by applying the same parallelism principle to arbitrary graph topologies. The gain is largest on RRG instances ($325\text{--}781\times$ over MH)

TABLE I
STATISTICAL COMPARISON OF THRML AGAINST MH AND STANDARD GIBBS. SIGNIFICANCE: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, NS = NOT SIGNIFICANT. r = RANK-BISERIAL EFFECT SIZE. $\uparrow$ = THRML ACHIEVES BETTER OUTCOME (LOWER ENERGY FOR ISING, HIGHER CUT FOR MAXCUT); $\downarrow$ = THE OTHER SAMPLER ACHIEVES BETTER OUTCOME. DIRECTION SHOWN ONLY WHEN SIGNIFICANT ($p < 0.05$).

Config	THRML vs MH		THRML vs Gibbs	
	sig	r	sig	r
ferro 50×50	ns	—	ns	—
ferro 70×70	ns	—	*** $\uparrow$	-0.31
ferro 100×100	ns	—	*** $\uparrow$	-0.48
antiferro 50×50	ns	—	* $\uparrow$	-0.11
antiferro 70×70	ns	—	*** $\uparrow$	-0.22
antiferro 100×100	** $\uparrow$	-0.21	*** $\uparrow$	-0.52
$\beta = 0.1$	ns	—	ns	—
$\beta = 0.5$	ns	—	*** $\uparrow$	-0.31
$\beta = 1.0$	ns	—	*** $\uparrow$	-0.25
$\beta = 2.0$	* $\downarrow$	0.13	** $\uparrow$	-0.19
RRG 900	ns	—	ns	—
RRG 1600	* $\uparrow$	-0.04	** $\uparrow$	-0.07
RRG 2500	ns	—	ns	—
MaxCut d3 900	* $\uparrow$	0.15	* $\uparrow$	0.16
MaxCut d3 1600	ns	—	ns	—
MaxCut d3 2500	** $\uparrow$	0.20	*** $\uparrow$	0.22
MaxCut d5 900	* $\uparrow$	0.17	ns	—
MaxCut d5 1600	* $\uparrow$	0.17	* $\uparrow$	0.14
MaxCut d5 2500	*** $\uparrow$	0.27	*** $\uparrow$	0.34

where full convergence is achieved, making the comparison unambiguous.

B. Solution Quality

THRML matches MH in solution quality across the majority of configurations and outperforms standard Gibbs on all large grid instances. This is a strong result: THRML forgoes the rejection step entirely, yet this does not lead to systematically worse solutions. The exception at $\beta = 2.0$ —where MH significantly outperforms THRML ($p = 0.028$, $r = 0.125$)—is consistent with the known limitation of Gibbs-based samplers at very low temperatures. At high β the energy landscape is sharply peaked, and MH’s rejection step allows it to avoid committing to slightly suboptimal block configurations. This is the same barrier-crossing mechanism identified by Roussel et al. in Restricted Boltzmann Machines [4], here observed on the Ising model.

Standard Gibbs consistently underperforms on large grid instances ($|r| = 0.30\text{--}0.52$ vs. THRML). Single-site random-order updates on a lattice produce strongly correlated samples because adjacent spins are updated sequentially rather than in parallel, preserving local correlations across steps. Block Gibbs eliminates this by updating all spins of a colour class simultaneously, which is why the quality gap between THRML and standard Gibbs grows with system size.

C. Convergence

The Gelman-Rubin diagnostics reveal an important limitation: all three samplers fail to satisfy $\hat{R} < 1.1$ on grid instances within the sample budget of 500 samples. This is not unique to

any one sampler—it reflects the intrinsic difficulty of mixing on large 2D lattices, where the correlation length grows with system size. Despite THRML’s superior ESS/s, the absolute number of effective samples at 100×100 ($\text{ESS} \approx 4.8$ per chain) remains low in absolute terms. Longer runs or larger sample budgets would be needed to achieve convergence on the largest grid instances.

In contrast, all three samplers converge fully on RRG instances ($\hat{R} \leq 1.001$), and MaxCut instances show near-convergence ($\hat{R} < 1.09$). The difference is structural: RRG and MaxCut graphs lack the long-range lattice order that makes the 2D grid difficult to mix on.

D. Effect of Graph Connectivity

The MaxCut degree sweep reveals that THRML’s quality advantage grows with graph degree: effect sizes are consistently larger at degree 5 than degree 3, and grow monotonically with problem size at degree 5. This is counterintuitive at first—higher degree increases the chromatic number, reducing the size of each parallel colour block and therefore THRML’s parallelism. However, denser graphs create more frustrated interactions, making the energy landscape more rugged and the sampling problem harder for all methods. In this regime, THRML’s faster decorrelation per step translates into a larger practical quality advantage, suggesting block Gibbs is particularly well suited to hard frustrated problems.

E. Implications for Thermodynamic Hardware

Jelinčič et al. project that the TSU will achieve performance parity with GPUs at roughly 10,000 times lower energy consumption [7]. Our results provide the first systematic empirical characterisation of the algorithmic behaviour of TSU-style block Gibbs across a range of problem classes. The finding that block Gibbs achieves equivalent or superior solution quality to MH at dramatically higher ESS/s supports the algorithmic premise of the TSU design. Importantly, these results represent a conservative lower bound on the benefit of thermodynamic sampling hardware: THRML runs on a conventional GPU, which is not architecturally optimised for stochastic computation. On dedicated TSU silicon, the same algorithmic efficiency advantage would compound with hardware-level energy savings, making the practical case stronger than the GPU simulation alone suggests. On the basis of these results, the TSU’s rejection-free block Gibbs approach appears well suited to MaxCut and ferromagnetic optimisation workloads. The $\beta = 2.0$ anomaly suggests caution for strongly ordered low-temperature regimes, where rejection-based methods retain a quality advantage.

F. Limitations

All experiments were conducted on a single NVIDIA RTX 3060 (6 GB VRAM). Results may differ on hardware with different memory bandwidth or parallelism characteristics. The sample budget (500 samples, 200 warm-up sweeps) is sufficient for RRG and MaxCut but insufficient for full convergence on large grid instances. Energy consumption was

not directly measured; ESS/s is used as a proxy for hardware efficiency, following the assumption that throughput advantage translates to energy advantage at equivalent solution quality [7]. Finally, all results reflect THRML running as a software simulation on GPU rather than on dedicated thermodynamic hardware; the efficiency advantages observed here do not include the projected hardware-level energy savings of the TSU.

VI. CONCLUSION

This thesis presented a systematic three-way comparison of THRML block Gibbs, standard single-site Gibbs, and Metropolis-Hastings on ferromagnetic Ising, anti-ferromagnetic Ising, and MaxCut problems across two graph topologies and a range of system sizes and temperatures.

RQ1 — Solution quality. THRML achieves solution quality statistically equivalent to MH across the majority of configurations, and significantly outperforms standard Gibbs on all large grid instances. On MaxCut degree-5, THRML consistently finds better solutions than both baselines at all system sizes, with effect sizes growing monotonically with problem size. The single exception is the very low temperature regime ($\beta = 2.0$), where MH’s rejection step confers a small but significant quality advantage. Overall, forgoing rejection does not lead to systematically worse solutions.

RQ2 — Scaling with topology and temperature. At high temperature ($\beta = 0.1$) all three samplers are statistically indistinguishable in quality; differences emerge as temperature decreases and the energy landscape becomes more rugged. On 2D grids, none of the samplers fully converge within the sample budget at large system sizes, reflecting the intrinsic difficulty of mixing on lattices. On random regular graphs and MaxCut instances, all three converge fully, and THRML’s quality advantage grows with graph degree, suggesting block Gibbs is particularly well suited to dense frustrated problems.

RQ3 — Computational efficiency. THRML achieves two to three orders of magnitude higher ESS/s than MH and standard Gibbs across all configurations. The gap is largest on random regular graphs (up to $781 \times$ over MH at 2,500 nodes), where full convergence makes the comparison unambiguous. This efficiency advantage holds even in cases where solution quality is equivalent, meaning THRML reaches the same quality with far fewer wall-clock seconds.

Societal implications. The energy cost of large-scale computation continues to grow, driven by machine learning workloads that rely heavily on sampling and optimisation. Thermodynamic computing proposes to address this by harnessing physical randomness rather than suppressing it. This work provides empirical evidence that the algorithmic foundation of the TSU—rejection-free block Gibbs sampling—achieves competitive or superior performance to established GPU-accelerated methods on the problem classes it targets. If the projected $10,000 \times$ energy efficiency of dedicated thermodynamic hardware [7] is realised, the combination of algorithmic and hardware efficiency could substantially reduce the energy

footprint of probabilistic inference and combinatorial optimisation workloads.

Future work. Three directions follow naturally from this study. First, longer runs on grid instances would establish whether THRML’s ESS/s advantage translates to full convergence given sufficient compute. Second, direct energy consumption measurement on TSU hardware would replace ESS/s as a proxy and allow a concrete comparison with GPU energy costs. Third, extending the benchmark to other NP-hard problem classes—such as graph colouring, satisfiability, or Bayesian inference networks—would establish how broadly the block Gibbs advantage generalises.

REFERENCES

- [1] N. Mohseni, P. L. McMahon and T. Byrnes, “Ising machines as hardware solvers of combinatorial optimization problems,” arXiv:2204.00276, 2022.
- [2] M. Weigel, “Performance potential for simulating spin models on GPU,” *J. Comput. Phys.*, arXiv:1101.1427, 2012.
- [3] M. Weigel and T. Yavors’kii, “GPU accelerated Monte Carlo simulations of lattice spin models,” arXiv:1107.5463, 2011.
- [4] C. Roussel, S. Cocco and R. Monasson, “Barriers and dynamical paths in alternating Gibbs sampling of Restricted Boltzmann Machines,” arXiv:2107.06013, 2021.
- [5] K. Y. Camsari, B. M. Sutton, and S. Datta, “p-Bits for probabilistic spin logic,” arXiv:1809.04028, 2019.
- [6] K. Y. Camsari et al., “Massively parallel probabilistic computing with Sparse Ising Machines,” arXiv:2110.02481, 2021.
- [7] V. Jelinčič et al., “An efficient probabilistic hardware architecture for diffusion-like models,” arXiv:2510.23972, 2025.
- [8] D. Melanson et al., “Thermodynamic computing system for AI applications,” *Nat. Commun.*, 2025. doi: 10.1038/s41467-025-59011-x.
- [9] Extropic, “THRML: Thermodynamic Hypergraphical Model Library,” 2025. [Online]. Available: <https://docs.thrml.ai>
- [10] J. Bradbury et al., “JAX: composable transformations of Python+NumPy programs,” 2018. [Online]. Available: <http://github.com/google/jax>
- [11] M. X. Goemans and D. P. Williamson, “Improved approximation algorithms for maximum cut and satisfiability problems using semidefinite programming,” *J. ACM*, vol. 42, no. 6, pp. 1115–1145, 1995.
- [12] A. Gelman and D. B. Rubin, “Inference from iterative simulation using multiple sequences,” *Statist. Sci.*, vol. 7, no. 4, pp. 457–472, 1992.
- [13] D. Brélaz, “New methods to color the vertices of a graph,” *Commun. ACM*, vol. 22, no. 4, pp. 251–256, 1979. doi: 10.1145/359094.359101.
- [14] C. J. Geyer, “Practical Markov chain Monte Carlo,” *Statist. Sci.*, vol. 7, no. 4, pp. 473–511, 1992. doi: 10.1214/ss/1177011137.